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→ Online Banking system

A. Purpose:-

The purpose of online Banking system is to provide customers with secure and convenient banking services through the internet.

The system allows customers to check their balance, transfer money, pay bills, view transaction history and manage their account. AI can also be used to analyze spending patterns and identify unusual transactions.

B. Users:-

The list of the users

are given below:-

- Customers
- Bank Employees
- Administrator
- Customer support staff
- Security officers

Customers:-

Customers use the system to manage their bank accounts, transfer money, pay bills and check transaction.

Bank Employees:

Employees manage customer accounts and provide banking services through authorized access.

Administrator:

The administrator manage users, system settings, database records and access permissions.

Customer support staff:

They help customers with login problems, transaction issues, and other banking queries.

Security offer:-

They monitor suspicious transactions and security alerts.

C- Stakeholders:

Stakeholders are people or organizations that are affected by the system or have an interest in its operation.

Customers:

They expect the system to be secure, reliable, fast and easy to use.

Bank Management:

They want the system to reduce operational costs and provide efficient banking services.

Bank Employees:

They use the system to perform their banking-related tasks.

IT/Software Team:

They develop, maintain, update and secure the system.

Regulatory Authorities:-

They require bank to follow applicable financial, privacy and security regulations.

D. S Functional Requirements:-

Functional requirements describe what the system should do.

- User Registration/login
- Balance checking
- Money transfer
- Bill payment
- Transaction History.

E-S Non-functional Requirements:-

Non-functional requirements

describe how well the system should work.

- Security
- Performance
- Availability
- Reliability
- Usability

Security:

The system must protect customer's personal and financial information

from unauthorized access.

e.g. Passwords protection, encryption, multi-factor authentication etc.

Performance:

The system should respond quickly when the customers perform common operations.

Availability:

The banking system should be available whenever customers need it, except during planned maintenance or unexpected outages.

Reliability:-

The system should process transactions correctly and consistently.

Usability:

The system should have a simple and understandable interface.

e.g. customers should easily find options such as Balance, Transfer, Bill etc.

I- RISKS:-

The risks of online Banking system are given belows:

- Cybersecurity Attack
- System Failure
- Incorrect Transaction Processing

Cyber-security Attack:

Hackers may try to gain unauthorized access to customer accounts or banking information.

Impact:- Financial loss, data exposure and loss of customer trust.

Mitigation: Strong authentication, encryption, monitoring, security testing.

System Failure:

A server, database or network problem could make banking services temporarily unavailable.

Impact:-

Customers may not be able to access accounts or perform transactions

Mitigations:

Back-ups, redundancy, monitoring disaster recovery and tested recovery procedure

Transaction Processing:-

A software or integration error could cause a transaction to be processed incorrectly.

Impact:-

incorrect balances or failed delay transfers.

Mitigations:

Transaction validation, database consistency control, transaction logs, testing.

9-3 Quality Attributes:-

Quality attribute describe important characteristics of a good software system.

1. Security:

The system should protect financial and personal information.

Eg. Only an authenticated and authorized customer should be able to access their account.

2. Reliability:

The system should perform its functions correctly and consistently.

Eg. When a transfer is completed, the system should correctly update the relevant account records.

3. Usability:

The system should be easy for customers to understand and operate.

Eg. The transfer process should clearly show the beneficiary.

amount and confirmation before
completing the transaction

H• where AI could be used: -

AI-Banking Assistant:

An AI assistance can
answer common banking questions
and guide user toward relevant
services.

Fraud/Unusual Transaction Detection: -

AI can analyze
transaction patterns and flag unusual
activity for further verification.

Spending Analysis:-

AI can categorized transaction
into areas such as Food, shopping,
Bills and Education and provide
a simple spending summary.

Personalized Financial Insights:-

AI can

identify spending patterns and provide general insights, such as showing which category has increased compared with an earlier period.

1. What could go wrong:-

False Fraud Alerts:-

AI may mark a genuine transaction as suspicious, which could inconvenience the customer.

Incorrect Information:-

If AI receives incomplete or incorrect data, it may provide an inaccurate answer to the customer.

So, the privacy issue, overdependence on AI, wrong financial suggestion are also involved in the wrong AI - use / incorrect use.